

PHIL 1002

Critical Thinking

Lecture 7A & 7B: Informal Fallacies



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In this and the next lecture

We will be completing **fallacies of clarity**

and introducing **fallacies of relevance**.

What are fallacies?

- spurious patterns of argumentation that create the illusion of cogency

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Recap: Fallacies of Clarity

Fallacies of clarity exist when the (mis)use of language gives an argument the *appearance* of strong support, or acceptable premises, where there is actually only weak or no support and/or unacceptable premise(s).

Common fallacies of clarity (or causes thereof):

- a) Equivocation
- b) Vagueness
- c) Emotionally loaded language
- d) Guilt by association
- e) Euphemism

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(c) *Slanting* with Emotionally loaded language

A related fallacy of clarity: "Slanting"



Slanting is the practice of including information in one's argument that is irrelevant to the conclusion, but serves to bias the audience or manipulate their emotions.

Example:

"The only thing that can make a practice morally wrong is that it causes suffering. Thus, there is nothing wrong with eating meat per se, provided the animals are raised humanely. Vegetarians, including Adolf Hitler, have been sorely mistaken about the moral status of eating meat."

Name of this particular fallacy: Guilt by Association

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(d) *Slanting* with Euphemism

Sample argument:

“The President’s primary responsibility is to keep the citizenry safe. Enhanced interrogations will yield information that could prevent future terrorist attacks on our soil. Hence, the President cannot ban enhanced interrogations without abdicating his responsibility to the citizenry.



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Before we move on...A new stipulative definition

Is there a word for someone who is more than an acquaintance

but less than a friend?

Now there is! What do you think that word is?

“Quend”

= someone who is more than an acquaintance but less than a friend.

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Part II: Fallacies of Relevance

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On Relevance

Recall the A.R.G. conditions for evaluating arguments (see Govier p. 87).

An argument is cogent when (and only when):

- 1) Its premises are **Acceptable**.
- 2) The premises are **Relevant** to the conclusion.
- 3) Considered together, the premises give **Good Grounds** (good reason) to accept the conclusion.

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Positive Relevance

Govier distinguishes three relations of relevance that can hold between two or more statements (A and B).

Positive Relevance:

The truth of A **counts in favour** of (or gives evidence for, or gives some reason to believe) that B is true.

E.g.

P1) The meteorologist predicts a 95% chance of rain tomorrow

C) It will **rain** tomorrow.



Positive relevance comes in **degrees**.

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Snapshot from 14.4.26

7 day forecast						Expand all
Today		Sunny.	18°	29°	No rain	▼
Tomorrow		Partly cloudy.	18°	27°	0–1mm	▼
Thursday		Mostly sunny.	16°	27°	No rain	▼
Friday		Possible shower.	17°	23°	0–1mm	▼
Saturday		Showers easing.	15°	21°	0–4mm	▼
Sunday		Partly cloudy.	11°	22°	No rain	▼
Monday		Possible shower.	12°	26°	0–1mm	▼

[← Hourly forecast](#)

[Past 72 hours →](#)

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Positive Relevance

Positive Relevance:

Example of weak positive relevance

P1) The meteorologist predicts a possible shower on Friday

C) It will rain on Friday.



As said, positive relevance comes in degrees.

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Negative Relevance

Govier distinguishes three relations of relevance that can hold between two or more statements (A and B).

Negative Relevance:

The truth of A counts against (or gives evidence against, or gives some reason not to believe) the claim that B is true.

P1. The meteorologist predicts a 95% chance of rain throughout the day tomorrow.

C. The weather will be clear and sunny tomorrow.



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Irrelevance

Govier distinguishes three relations of relevance that can hold between two or more statements (A and B).

Irrelevance:

The truth of A counts neither in favour of nor against the claim that B is true.

P1. The meteorologist predicts a 95% chance of rain throughout the day tomorrow

C. There is a red car parked in the student carpark.



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Fallacies of Relevance

“Fallacies of relevance are mistakes in argument involving irrelevance [or inflation of relevance] of one or more premises to the conclusion” (Govier, p. 154)

Question:

Can a premise that is positively relevant to a conclusion commit a fallacy of relevance?

YES!!!

in some cases, the problem may be that the offending premise gives *the appearance of much greater* relevance than it actually has.

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Fallacies of Relevance

- In yet other cases, the context may invite us to insert a missing premise so that, taken together, the support is strong.
- However, if the inserted premise is not acceptable, then the problem has not disappeared.
- Instead of failing the 'R' condition, it fails the 'A' condition of the ARG criteria.
- In a broader sense one commits a **fallacy of relevance** by **omitting information that's relevant to the conclusion and hence the strength of the argument**.
- That is the sense of 'relevance' we'll be talking about when we talk about fallacies of relevance.

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Fallacies of Relevance

Two fallacies of relevance

a) False dichotomy

b) Straw man

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(a) False Dichotomy

An argument is guilty of positing a **false dichotomy** when one of its premises asserts that a pair of options are exhaustive when, in fact, the options are not exhaustive—that is, there are other options available.

E.g.

P1. Each person is either rich or poor.

P2. Marvin is not rich.

C. Marvin is poor.

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False dichotomy

Polarised thinking in places where it should not be polarised.

1. **Either you are my friend or you are my enemy**

2. You are not my friend

Therefore:

3. You are my enemy



1. **Either you rode a bike to school or you drove**

2. You did not ride to school

Therefore:

3. You drove



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Statements involving contradictories

A false dichotomy involves a disjunctive 'either/or' statement that purports to cover **all** the relevant alternatives when it actually leaves some of them out.

It is useful to know that there are some statements involving disjuncts that **do** cover all the alternatives.

These statements involve contradictories.

Contradictories have opposite truth values.

- Each logically excludes the other:

- cannot be both true

- cannot be both false

A is true iff B is false / B is true iff A is false.

[note these are logically equivalent]

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Examples of statements involving contradictories

He is either a UWA student or he is not.

She is either pregnant or she is not.

Morally speaking, that action is either permissible or impermissible.

Either you see it or you don't.

Either PROSH is happening this week or it is not.



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Statements involving contraries

Statements involving contraries have disjuncts that do not cover all the logical alternatives.

With contraries the disjuncts *cannot both be true* but *can both be false*.

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Examples of statements with contraries

Either you are my friend or my enemy

Either you rode a bike to school or you drove

Her natural hair colour is either blonde or black

He is either a vegan or a meat-eater.

You're either a bundle of joy or a pit of gloom!

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Statements with contraries

False dichotomies can occur when statements with contraries masquerade as statements with contradictories, pretending to cover all the alternatives when they do not.

Question:

Are statements with contraries the only kind that may be involved in a false dichotomy?

Reminder: with contraries, the disjuncts can both be false but cannot both be true.

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Does this argument commit the fallacy of false dichotomy?

P1 The disjunctive premise in a false dichotomy expresses a statement that involves either a contrary or a contradictory pair of alternatives.

P2 The disjunctive premise in a false dichotomy does not involve a contradictory pair of alternatives

C The disjunctive premise in a false dichotomy involves a contrary pair of alternatives

Depends on whether P1 is true!

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In a false dichotomy

Sometimes both alternatives in a false dichotomy presented can be true (as well as both false) so they are **not contraries**

Consider this example:

P1: Either it is sunny or it is raining

P2 It is not sunny

C It is raining

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(a) False Dichotomy: trickier cases

Note: Often, a premise asserting a false dichotomy is not explicitly stated in the argument, but is instead implicit or “suppressed”.

E.g.

P1. David does not support *laissez-faire* capitalism.

C. David is a socialist.

Implicit premise:

P2. Either a person supports *laissez-faire* capitalism, or else they are a socialist.

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Trump Drives By Well-Wishers, Raising Concerns About Health of Secret Service Agents in His Car

“Mr. Cortes [a senior adviser to Mr. Trump’s campaign] outlined a false choice, suggesting that the alternative to holding large events without masks was to remain “hermetically sealed” in the White House.”

Source: The New York Times, last updated Nov 18, 2020



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Reconstruction of Mr Cortes' reasoning

P1 Either its OK to remain hermetically sealed in the White House or it's OK to hold large campaigns without masks

P2: It's not OK to remain hermetically sealed in the White House

C: It's OK to hold large campaigns without masks

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A more complex example

Interviewer:

"Is there a [moral] problem about genetic engineering, in applying it to human beings?"

Watson (DNA man):

"Humans are always intervening"

Suppressed reasoning:

[P1: **Either we should reject all intervention or accept that GE is morally OK**]

P2 **We should not reject all intervention**

C **GE is morally OK**

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GWB's State of Union address



“Every nation, in every region, now has a decision to make.
Either you are with us, or you are with the terrorists”

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MORE Fallacies of Relevance

- False Dichotomy
- Straw Man
- Genetic Fallacy
- Ad Hominem
- Appeal to Ignorance

Take note!

{ Govier discusses other
fallacies in Ch. 6;
Know them!

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(b) Straw Man



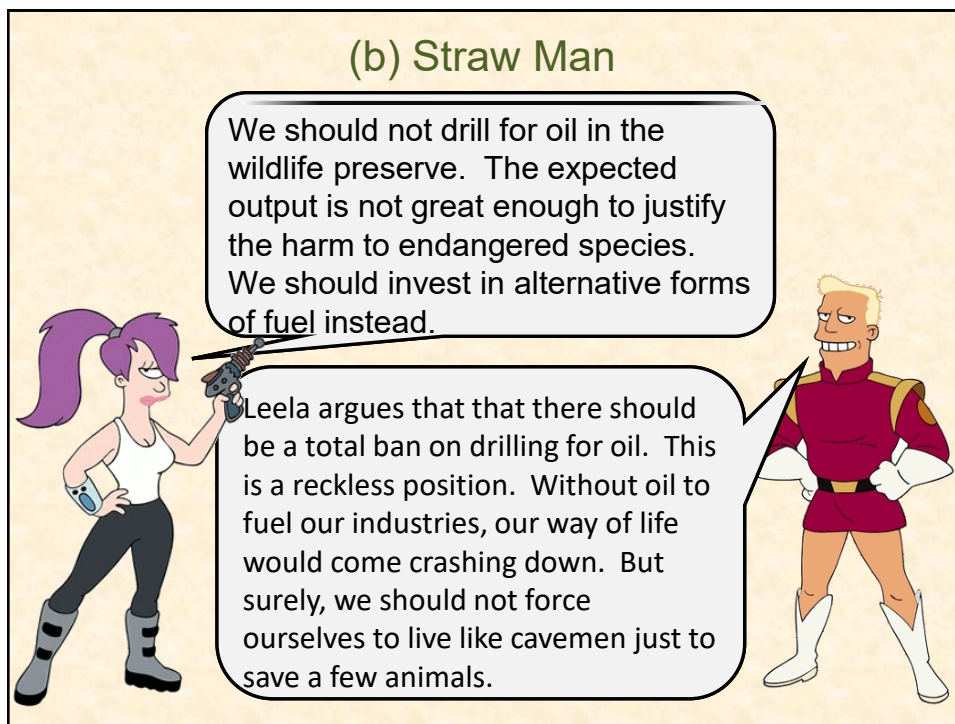
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(b) The Straw Man Fallacy

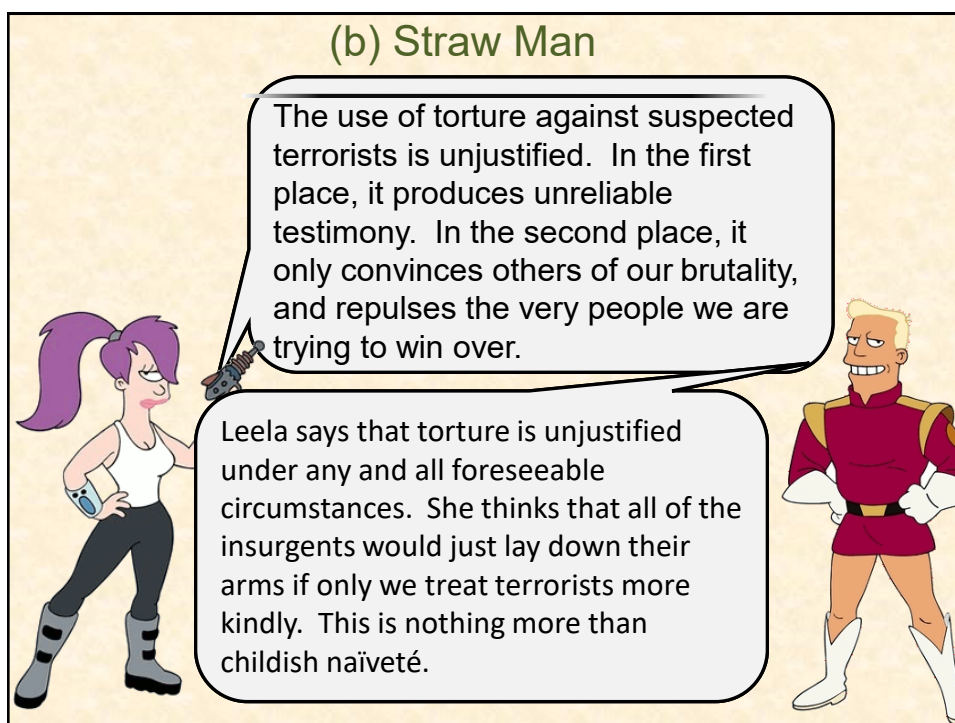
“The **straw man** fallacy is committed when a person misrepresents an argument, theory, or claim, refutes that misrepresentation, and then claims to have refuted the position [that has been] misrepresented” (Govier, p. 155).

- Typically, the fallacy involves recasting an opponent's viewpoint in a more extreme and/or less nuanced way.
- Constructing an argument against a straw man is a failure of relevance, since one's argument is not actually directed against the claim that it presents itself as being directed against.

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Sometimes more than one fallacy can exert itself at a time

Interviewer:

"Is there a problem about doing [the sort of] experimental work on animals [that you are doing]?"

Watson (DNA man):

"Animal Rights people want us never to touch animals."

Strawman – portrays animal rights people as uniformly objecting to all animal experimentation

False Dichotomy: P1 Either we should ban all animal experimentation Or accept their use in genetic engineering

P2 Experiments on animals should not be banned

C It is morally OK to use animals for genetic experimental work

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The Genetic Fallacy

Consider the following exchange:

God exists. For God is defined as the greatest conceivable being. Great-making qualities will include not only perfect power, knowledge and love, but also existence.

I reject your conclusion. You only believe that God exists because your parents drilled this belief into you at a young age. In addition, you cling to this belief out of fear of your own mortality.



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The Genetic Fallacy (not discussed in Govier)

The **genetic fallacy** is committed when claims about *how a given belief (or institution) originated* are asserted as reasons for or against thinking that the belief (or institution) is justified or warranted.

...the G.F.'s clearest violations occur when someone else gives evidence or argument in favour of a belief, and the another offers a counterargument that (fallaciously) highlights the dubious origins of the first belief without engaging with the reasons offered in defense of it.

...but it can also occur in direct justification for a claim (such as defending an institution or practice)

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The Genetic Fallacy

E.g.:



Some people say we ought not to go to war with Spheron I. But it is just their own cowardice that causes them to think this. Hence, we should go to war with Spheron I.

A general point:

Facts about the **origin** of a belief or institution are often not directly relevant to facts about the **merit** or justification of that belief or institution.

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The Genetic Fallacy

Committing the G.F. in an attempted defense of an institution or practice:

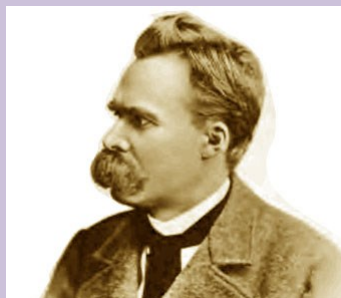


"When the human species evolved, our ancestors lived in hunter-gatherer societies. The men left the encampments to hunt for meat, while the women stayed behind and looked after the children. It was in this sort of circumstance that human families first developed. Consequently, a woman's proper place within a family is always at home with the children."

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The Genetic fallacy

"Friedrich Nietzsche's view on women is irrational and should not be taken seriously. He could not sustain healthy relations with most of the women in his life."



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Genetic fallacy



“Red meat. We were meant to eat it”

<https://www.youtube.com/watch?v=hNWB6Vq18xc>

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Ad Hominem

Argumentum Ad Hominem:

“argument against the man.”

Given the outbreak of violent crime, the government ought to bring back capital punishment. Only the death penalty will deter murderers.

I'm not surprised to hear Philip express such a viewpoint. He is a rather dull person. He doesn't wash much either.



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Ad Hominem

Argumentum Ad Hominem:

“argument against the man.”

“...is one in which a critic [aiming to rebut an argument] attacks a person instead of arguing against the claims that person has put forward.”

Sometimes it involves attacking the person's **character**, appearance, intelligence, or behavior.

Sometimes it involves attacking the person's **motives**.

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Ad Hominem

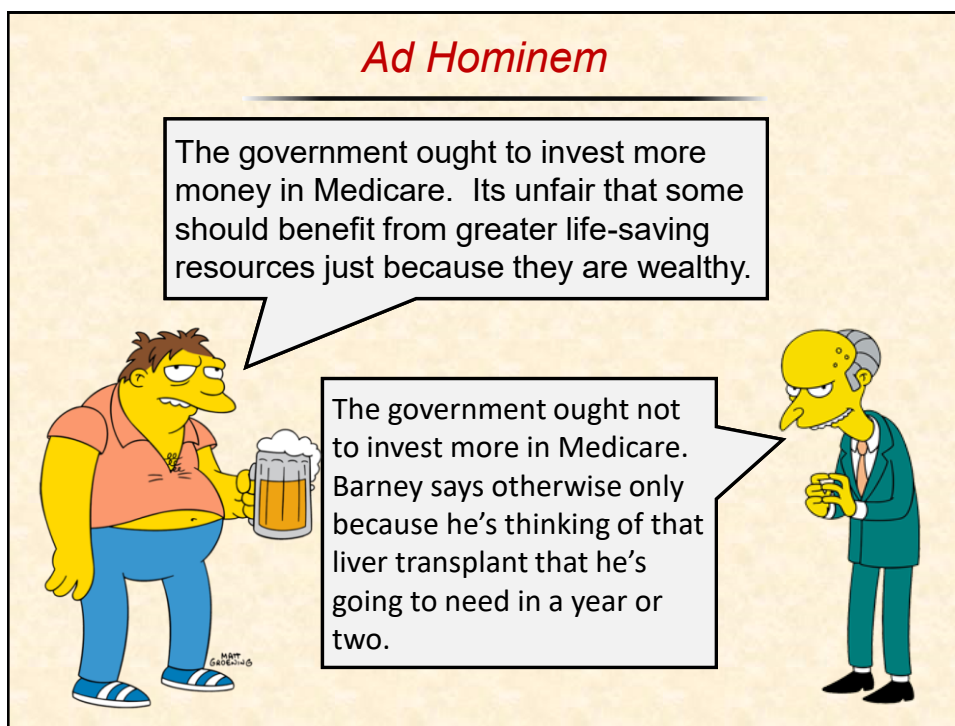
The recession is being prolonged by high taxes on the wealthiest individuals. A big tax cut for those earning over \$1 Million would encourage investment and jump- start the economy.



Don't listen to Mr. Burns. He's only in favor of the tax cut because he's a millionaire and stands to benefit from it personally.



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Example of Ad hominem and slanting using emotive language:

Background: Newscorp columnist Andrew Bolt is using Greta's autism as a reason to ignore her arguments for paying heed to climate change. Reported in the Guardian 1.8.19



The widely read Herald Sun columnist and Sky News commentator used his significant platform to take aim at the 16-year-old campaigner, **dismissing her followers as members of a cult** and disparaging her decision to **sail across the Atlantic** in a high-speed racing yacht to attend UN climate summits in the US and Chile.

“Thunberg has announced she’s finally going to the United States, the last bastion of the heathen, to preach the global warming faith to the Americans,” Bolt wrote. “Of course, she’s going by racing yacht, because she refuses to fly and heat the planet with an aeroplane’s global warming gasses.”

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Example of Ad hominem and slanting using emotive language: Background:
Newscorp columnist Andrew Bolt is using Greta's autism as a reason to ignore her arguments for paying heed to climate change.
 Reported in the Guardian 1.8.19

The highly personal character assassination published in Rupert Murdoch's tabloids repeatedly referred to Greta's mental health, saying she was "deeply disturbed", "freakishly influential" and "strange".

"I have never seen a girl so young and with so many mental disorders treated by so many adults as a guru," Bolt wrote.

"Far more interesting is why so many adults - including elected politicians, top business leaders, the Pope and journalists - treat a young and strange girl with such awe and even rapture.

"Her intense fear of the climate is not surprising from someone with disorders which intensify fears."

Bolt even described Greta's younger sister as having "a spectacular range of mental issues".

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Tu Quoque fallacy

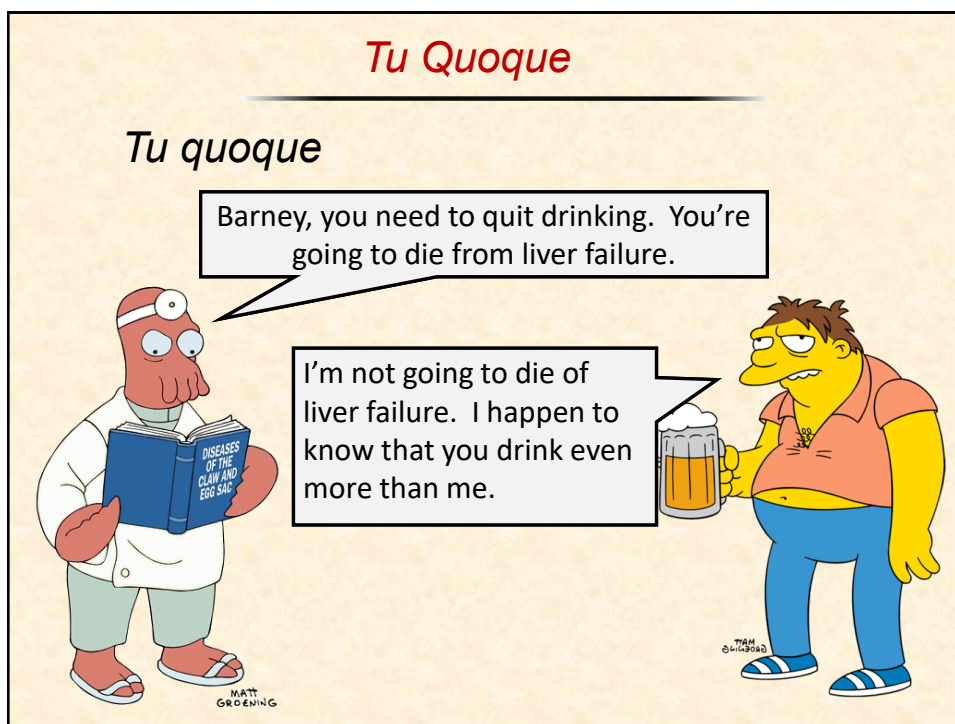
A sub-type of the Ad Hominem fallacy:

Tu quoque: "You too."

- A **Tu quoque** fallacy is "...is committed when it is alleged that an arguer has a fault similar to the one he or she is criticizing, and then, on the basis of that allegation, it is inferred that the criticism can be dismissed."

(Govier, 6th Ed. P. 203)

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Legitimate Ad Hominems

Three cases where *Ad hominem*s (or something much like it) are legitimate:

- (1) The person's character or motives (etc.) are relevant to the point being made.
(e.g. Someone is being paid to argue how good a product is...)
- (2) We are being asked to accept a claim on the basis of someone's authority or testimony, and we need to establish their credibility.
(e.g. In court, Jones says he saw Smith at the scene of the crime, but Jones has been convicted of perjury more than once!)



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Legitimate *Ad Hominems*

Three cases where *Ad hominem*s (or something much like it) are legitimate:

- (3) To show that someone's beliefs are inconsistent, and that they are therefore rationally required to abandon at least one of those beliefs.

Marg: "Using live animals for scientific research should be banned. It is painful to the animals and the research could be done in other ways."

Barney: "But you eat meat; and you don't worry that the animals suffer while being killed. Moreover, you could get your protein in other ways."



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Appeal to Ignorance

There is no evidence that P is true

Therefore

P is false

There is no evidence that P is false

Therefore

P is true

Consider:

P = 'angels exist'



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Appeal to Ignorance

- We may commit the fallacy of **appealing to ignorance** when we try to argue that a claim is true [or false] on the grounds that **we do not know** (or do not have evidence) that the claim is false [or true]



Angels exist. After all, there is no evidence that they do not exist.

There is no life on other planets. After all, no one has ever found evidence of any life on other planets.

Slogan: Absence of evidence is not evidence of absence!

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But note!

- Sometimes, however, an appeal to ignorance is **not** fallacious.
- Suppose that someone searches for your car at the bottom of the river using scuba divers and sonar and nothing shows up.
- Or we look at James Oval and say 'there's no evidence of an elephant on the Oval, so there's no elephant on the Oval'
- In such a case, absence of evidence **is** evidence of absence.
- What is the difference between these cases and that of the angels or extra-terrestrials?*

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Non-fallacious Appeal to Ignorance

- When is it not fallacious?

“If a responsible, appropriate, and careful search for some entity, X, has not turned up any evidence of X, then there is reason to believe that X is not there to be found.”

(Govier: p. 165)

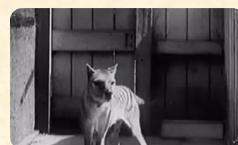
To this we can add: If X *had* been present, evidence of X *would* have been found.



Your car is not at the bottom of the lake. Scuba divers and sonar have searched it and found nothing. If your car had been at the bottom of the lake, they would have found it.

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Non-fallacious Appeal to Ignorance: example



P There is no evidence that there are Tasmanian Tigers in Tasmania now.

C There are no Tasmanian Tigers in Tasmania now.

Were Tasmanian Tigers still around isn't it likely that we would have evidence for their existence? Yes. Is it likely they would be around and not detected? No.

So the lack of evidence, in this case, supports the claim that there are no Tasmanian Tigers in Tasmania.

Lack of evidence for X may be relevant when:

- people have carefully looked for evidence of X and have failed to find it **AND**
- had X been around, it is likely that evidence for it would have been found

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'Tasmanian tiger' footage not of a thylacine but a pademelon, experts say

Continuing to search for [Tasmanian Tigers] is a waste of money, according to Mike Archer from UNSW Sydney.

"If there were thylacines in Tasmania, there'd be roadkill."



Pademelon

ABC News, Posted Wednesday 24 FebFebruary 2021 at 11:16am

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Non-fallacious Appeal to Ignorance: Burden of Proof

Burden of proof: Another exception: in some contexts we seem to think that one side has “the burden of proof”. In this case, we think that there is a default position to take on an issue, and in the lack of positive evidence we are justified in holding the default position. (In some cases, an argument can be given for why the default position deserves its status).



[uttered the late 1950s] There's no positive evidence that Thalidomide is dangerous. No one's ever tested the drug, so there's no reason to worry that it is dangerous.

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Non-fallacious appeal to ignorance: Burden of proof

There's no evidence that Thalidomide is dangerous
therefore

Thalidomide is safe

Versus

There's no evidence that Thalidomide is safe
therefore

Thalidomide is dangerous.

The default position we practically take is the **second** inference.

The **burden of proof** lies on the side of the **first inference** to SHOW that there's no evidence that the drug is dangerous.

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Non-fallacious appeal to ignorance: Burden of proof

There's no evidence that he is guilty
therefore

He is innocent

Versus

There's no evidence that he is innocent
therefore

He is guilty.

The default position we practically take is the **first** inference.

The burden of proof lies on the side of the **second** inference to SHOW beyond reasonable doubt that evidence for the defendant's innocence is lacking.

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Appeals to Ignorance: consider this argument. Is it fallacious?

P. There is no positive evidence that angels exist
C*. There is no good reason to believe that angels exist

This is OK!

- **There is a DIFFERENCE between:**

P1 There is no positive evidence that P

C. It is false that P

and:

P1 There is no positive evidence that P

C* There is no good reason to believe that P

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End

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